Beaches or Books: Tourism and Human Capital Accumulation*

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Abstract

[Abstract placeholder.]

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1 Introduction

Increases in human capital are a key mechanism by which countries are able to raise productivity, grow their economies, and improve living standards. It is therefore important to understand the factors that determine the level of investment by households in the education of their children. Existing work has demonstrated the important role played by factors such as income constraints, expected returns to education, and contemporaneous shocks to economic opportunities (Shah and Steinberg 2017; Atkin 2016). The effect of positive economic shocks on schooling decisions are known to depend on whether the income effect of suddenly having greater available resources to spend on education dominates, or the substitution effect, of the increased opportunity cost of schooling in light of greater income opportunities in the short-term is dominant. While there has been significant work conducted studying the effects of such shocks to agriculture and manufacturing, less attention has been paid to the role of positive shocks to service sector industries on schooling decisions. This paper addresses this gap in the literature by studying the effect of specialization in tourism services on human capital investment in Jamaica.

This paper answers the question: What is the impact of increases in the size of a local tourism industry on the decisions by households on how much to invest in education. Additionally, this paper studies how much the impacts of increases in the intensity of tourism vary depending on the stage of education. I employ two decades of rich tourism exit surveys spanning the year 2001-2021, which provide a detailed picture of tourist spending across Jamaican communities. I combine these with data from the Jamaica Survey of Living Conditions (JSLC) a representative household survey that provides thorough information on the spending by households on the education of individual children, as well as demographic insights such as the educational attainment and employment of parents. Linking households and tourism spending across spatially consistent units, I use a shift-share instrumental variable account for the likely endogeneity of development area tourism levels. I leverage exogenous variation in the number of tourists visiting Jamaica from specific regions of the world from year to year, and the heterogeneous preferences of tourists from various origin countries for where they choose to stay in Jamaica.

I find that one-year lagged tourism levels produce no impact on the enrollment or attendance levels of students currently in school. These effects are consistent across all levels of schooling.

2 Context and Background

2.1 Jamaica and Tourism

[Context and background placeholder.]

Jamaica is an upper-middle-income island nation of approximately three million people located in the Caribbean Sea. Tourism is the foundation of Jamaica's development strategy and its single largest foreign-exchange earner. The sector employs roughly one in four Jamaicans, either directly or indirectly, and tourism revenues as a share of GDP have consistently exceeded 20 percent in recent years. The country specializes in beach-based, all-inclusive resort tourism, drawing visitors primarily from North America and Europe.

Tourism activity is highly heterogeneous across Jamaica's development areas — the spatial units used by the Statistical Institute of Jamaica (STATIN) and the Social Development Commission (SDC) to group communities by geographic and socioeconomic characteristics. Resort towns such as Montego Bay, Negril, and Ocho Rios receive the vast majority of tourist overnight stays, while inland and rural development areas have much lower exposure to tourist spending. This geographic variation, combined with year-to-year variation in the origins of tourist arrivals, provides the identifying variation used in this paper.

2.2 Human Capital, Household Income, and Tourism

3 Data

[Data section placeholder.]

Jamaica Survey of Living Conditions (JSLC)

The primary household data source is the Jamaica Survey of Living Conditions (JSLC), a nationally representative rotating cross-sectional household survey conducted annually by the Statistical Institute of Jamaica and the Planning Institute of Jamaica. The JSLC collects detailed information on household consumption expenditures, income, employment, demographics, and — critically for this paper — a dedicated education module covering school enrollment, attendance, grade level, exam outcomes, and school-related expenditures. I use waves spanning 2006–2019, excluding years 2005, 2011, and select survey years that lack key education variables.

The education module of the JSLC records, for each household member of school age, the type of school currently attended, the number of days sent to school in the reference week, reasons for any absences, and school-related expenditures on tuition, transport, uniforms, books, and lunch. For individuals no longer currently enrolled, the survey records the last grade and school type completed, years of schooling, and the reason for stopping attendance. National examination outcomes — specifically, whether the respondent passed mathematics and English at the CSEC level — are also recorded.

Ministry of Tourism Exit Surveys

Tourist expenditure data come from the Ministry of Tourism (MOT) of Jamaica, which administers an exit survey to departing tourists at major ports of departure. These surveys record detailed information on tourist spending, country of origin, length of stay, accommodation type, and the geographic areas of the island visited. I use waves of the MOT exit survey spanning 2001–2019.

Tourist spending is linked to development areas using information on accommodation location. I construct a measure of total tourist accommodation expenditures attributed to each development area in each year, $\mathrm{DA\_EXP}_{dt}$, which is the primary continuous treatment variable in the analysis.

Shapefiles and Geographic Matching

Development area boundaries from STATIN shapefiles are used to spatially match JSLC households to their development area and to allocate tourist spending to development areas. The JSLC reports development areas directly for the years covered, facilitating this linkage.

4 Empirical Approach

4.1 The Shift-Share Instrumental Variable Approach

The central empirical challenge is that the level of tourism activity in a development area may be correlated with local economic conditions that independently affect household education decisions. To address this endogeneity, I employ a shift-share (Bartik) instrumental variable strategy.

The instrument is constructed by interacting aggregate "shifts" — year-to-year changes in the number of tourists from each origin country o arriving in Jamaica, ΔN_{ot} — with predetermined "shares" capturing each development area d's historical exposure to tourists from origin country o, s_{od}. The Bartik instrument is:

$$B_{dt} = \sum_{o} s_{od} \cdot \Delta N_{ot} \tag{1}$$

where s_{od} is the share of development area d's tourist accommodation expenditures attributable to origin country o in a base year, and ΔN_{ot} is the national-level change in arrivals from origin country o in year t. The identifying assumption is that the national-level shifts in origin-country arrivals are orthogonal to local development-area shocks to education outcomes, conditional on development area and year fixed effects.

The baseline estimating equation is:

$$Y_{idt} = \alpha + \beta\, \widehat{\mathrm{DA\_EXP}}_{dt} + \mathbf{X}'_{idt}\gamma + \delta_d + \lambda_t + \varepsilon_{idt} \tag{2}$$

where Y_{idt} is an education outcome for individual i in development area d in year t; $\widehat{\mathrm{DA\_EXP}}_{dt}$ is the fitted value of local tourism accommodation expenditures from the first stage using B_{dt} as the excluded instrument; $\mathbf{X}_{idt}$ is a vector of individual and household controls; δ_d and λ_t are development area and year fixed effects; and ε_{idt} is a disturbance term clustered at the development area level.

4.2 Identification Checks

[Identification checks placeholder.]

The validity of the shift-share instrument rests on two conditions. First, the shifts (origin-country arrival changes) must be plausibly exogenous to local Jamaican development-area conditions. Year-to-year variation in Canadian, American, and European tourist arrivals is driven primarily by macroeconomic conditions, exchange rates, and travel costs in origin countries — factors that are unlikely to directly affect school attendance or exam performance in specific Jamaican development areas. Second, the shares must be predetermined relative to the shocks; I use base-year shares constructed from the earliest available exit survey years to ensure this.

I test the exclusion restriction in several ways, following Goldsmith-Pinkham et al. (2020). I verify that the instrument is uncorrelated with pre-trend changes in education outcomes, and I test for balance across pre-determined household characteristics.

4.3 Interpreting the Coefficient

The coefficient β in equation (2) is interpreted as the causal effect of an additional million USD in development area tourism revenues on the education outcome Y, for households in that development area. This is a local average treatment effect (LATE) for households in development areas with variation in tourism revenues driven by origin-country arrival shocks.

5 Results

5.1 Baseline Findings: School Attendance and Enrollment

[Results placeholder.]

Table 1: School Enrollment (Ages 5–19), IV Estimates

Dependent Variable:	ENROLLED_IN_SCHOOL		
Model:	(1)	(2)	(3)
Variables			
Tourism exp., lag 1 (DA, M USD)	-6.72×10^{-5}	-6.73×10^{-5}	0.0008
	(0.0004)	(0.0004)	(0.004)
Female	0.006*	0.006*	0.014
	(0.003)	(0.003)	(0.025)
Age	-0.038***	-0.038***	-0.043***
	(0.001)	(0.001)	(0.003)
No. enrolled siblings (HH)		-0.0002	
		(0.002)	
Fixed-effects			
DEV_AREA	Yes	Yes	Yes
YEAR	Yes	Yes	Yes
Fit statistics			
F-test (1st stage), DA_EXP_lag1	1,028.8	1,030.2	44.088
Observations	13,518	13,518	861
R^2	0.25524	0.25524	0.31733

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 2: Reason for Non-Enrollment (Ages 5–19, Not Currently Enrolled), IV Estimates

Dependent Variables:	DROPOUT_MONEY	DROPOUT_FAMILY	DROPOUT_NO_INTEREST	DROPOUT_PREGNANT
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 1 (DA, M USD)	0.0003	0.0002	-0.0001	0.0003
	(0.0004)	(0.0003)	(0.0005)	(0.0005)
Female	-0.031***	-0.004	-0.040***	0.080***
	(0.011)	(0.009)	(0.010)	(0.011)
Age	-0.004	-0.005**	-0.005	-0.003
	(0.004)	(0.002)	(0.003)	(0.002)
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_lag1	172.62	172.62	172.62	172.62
Observations	1,637	1,637	1,637	1,637
R^2	0.08282	0.04960	0.04458	0.07234

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: School Attendance Levels (Enrolled Sample), IV Estimates

Dependent Variables:	SCHOOL_DAYS_SENT_PER_WEEK		KEPTFROMSCHOOL_ANY	
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 1 (DA, M USD)	-0.002	-0.002	0.0004	-0.004
	(0.007)	(0.007)	(0.0004)	(0.007)
Female	0.035	0.035	0.001	0.055***
	(0.111)	(0.111)	(0.005)	(0.020)
Age	-0.029***	-0.030***	-0.007***	-0.009**
	(0.011)	(0.011)	(0.0008)	(0.004)
No. enrolled siblings (HH)		-0.034		
		(0.077)		
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_lag1	861.62	862.81	862.14	42.975
Observations	11,412	11,412	11,881	756
R^2	0.03428	0.03435	0.22951	0.31536

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: Reasons for School Absence (Occasional or Frequent, Annual; Enrolled Sample), IV Estimates

Dependent Variables:	SCHOOL_KEPTYEAR_ANY_OCC_OR_FREQ	SCHOOL_KEPTYEAR_MONEY_OCC_OR_FREQ	SCHOOL_KEPTYEAR_TRANSPORT_OCC_OR_FREQ	SCHOOL_KEPTYEAR_ILL_OCC_OR_FREQ	SCHOOL_KEPTYEAR_WORK_OCC_OR_FREQ	SCHOOL_KEPTYEAR_MONEY_OCC_OR_FREQ
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp., lag 1 (DA, M USD)	-0.002**	0.0002	0.0002***	-0.002**	1.68×10^{-5}	-0.008
	(0.0009)	(0.0007)	(5.63×10^{-5})	(0.0008)	(2.37×10^{-5})	(0.015)
Female	-0.026***	-0.017**	-7.3×10^{-5}	-0.001	-0.0004	-0.015
	(0.006)	(0.007)	(0.001)	(0.004)	(0.0004)	(0.042)
Age	-0.002*	0.002*	0.0004***	-0.004***	3.76×10^{-5}	0.004
	(0.001)	(0.0009)	(0.0001)	(0.0005)	(5.38×10^{-5})	(0.006)
Fixed-effects						
DEV_AREA	Yes	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
F-test (1st stage), DA_EXP_lag1	862.14	862.14	862.14	862.14	862.14	42.975
Observations	11,881	11,881	11,881	11,881	11,881	756
R^2	0.11163	0.06948	0.00209	0.02901	0.00624	0.10541

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

5.2 School Expenditures

[Results placeholder.]

Table 5: School Expenditure by Category, Level (Enrolled Sample, Real 2024 USD), IV Estimates

Dependent Variables:	SCHOOL_TUITION_NOBOOKS	SCHOOL_BOOKS_EXPENSES	SCHOOL_TRANSPORT_EXPENSES	SCHOOL_UNIFORM_EXPENSES	SCHOOL_LUNCHSNACKS_EXPENSES
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 1 (DA, M USD)	-0.181	-0.237***	0.610	-0.527	0.519
	(0.332)	(0.089)	(0.666)	(0.316)	(0.835)
Female	-3.93	5.57***	25.3***	-8.15***	15.4**
	(6.27)	(1.89)	(7.98)	(1.90)	(6.63)
Age	-7.26***	1.15***	32.0***	2.68***	29.4***
	(0.797)	(0.247)	(2.48)	(0.270)	(1.28)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag1	430.30	633.61	636.95	655.96	631.13
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.04089	0.06375	0.07658	0.02449	0.11447

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 6: School Expenditure by Category, Log Outcomes (Enrolled Sample), IV Estimates

Dependent Variables:	LOG_SCHOOL_TUITION_NOBOOKS	LOG_SCHOOL_BOOKS_EXPENSES	LOG_SCHOOL_TRANSPORT_EXPENSES	LOG_SCHOOL_UNIFORM_EXPENSES	LOG_SCHOOL_LUNCHSNACKS_EXPENSES
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 1 (DA, M USD)	-0.005	-0.039*	0.023	-0.012	0.013
	(0.019)	(0.021)	(0.022)	(0.010)	(0.011)
Female	-0.003	0.657***	0.990***	0.058	-0.084
	(0.142)	(0.210)	(0.221)	(0.137)	(0.110)
Age	-0.551***	-0.323***	1.13***	0.024	0.185***
	(0.035)	(0.037)	(0.039)	(0.019)	(0.020)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag1	430.30	633.61	636.95	655.96	631.13
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.08508	0.05588	0.13410	0.02069	0.02585

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

5.3 Educational Attainment and Exam Performance

[Results placeholder.]

Table 7: Examination Attainment: At Least Level X (Ages 5–19), IV Estimates

Dependent Variables:	EXAM_ATLEAST_GNAT	EXAM_ATLEAST_CXC_BASIC	EXAM_ATLEAST_CXC_GEN		MATHENG_BOTH
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 1 (DA, M USD)	-6.98×10^{-5}	-0.0002	-6.6×10^{-5}	0.0005	5.36×10^{-5}
	(0.0002)	(0.0001)	(0.0001)	(0.0009)	(0.0001)
Female	0.031***	0.030***	0.025***	0.018*	0.011***
	(0.004)	(0.004)	(0.004)	(0.010)	(0.002)
Age	0.021***	0.019***	0.016***	0.008***	0.007***
	(0.0008)	(0.0009)	(0.0010)	(0.002)	(0.0008)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag1	1,028.8	1,028.8	1,028.8	44.088	1,028.8
Observations	13,518	13,518	13,518	861	13,518
R^2	0.14510	0.13709	0.11767	0.10563	0.05217

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8: Upper Secondary and Tertiary Attainment (Ages 14–25), IV Estimates

Dependent Variables:	EXAM_ATLEAST_ALEVEL	EXAM_ATLEAST_DEGREE	CURRENTLY_IN_COLLEGE		
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 1 (DA, M USD)	-0.0003	-9.35×10^{-6}	-0.0004**	-0.0005**	-0.0009
	(0.0002)	(0.0001)	(0.0002)	(0.0002)	(0.001)
Female	0.023***	0.007***	0.029***	0.028***	0.021*
	(0.004)	(0.002)	(0.004)	(0.005)	(0.012)
Age	0.009***	0.003***	0.005***	0.006***	0.003
	(0.001)	(0.0006)	(0.0007)	(0.0010)	(0.002)
No. enrolled siblings (HH)				-0.005**	
				(0.002)	
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag1	832.65	832.65	832.65	652.46	31.077
Observations	9,706	9,706	9,706	8,116	429
R^2	0.05311	0.02438	0.03125	0.03236	0.18708

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 9: Skills Program Enrollment and Completion (Ages 5–19), IV Estimates

Dependent Variables:	ENROLLED_SKILLED_PROGRAM		DIPLOMA_SKILLS_PROGRAM	
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 1 (DA, M USD)	-6.4×10^{-5}	-6.23×10^{-5}	0.0004	0.416
	(0.0001)	(0.0001)	(0.0004)	(9.53)
Female	0.004	0.003	-0.004	1.35
	(0.002)	(0.002)	(0.012)	(32.3)
Age	-0.0006	-0.0006	0.015***	0.348
	(0.0007)	(0.0007)	(0.004)	(7.63)
No. enrolled siblings (HH)		-0.001*		
		(0.0005)		
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_lag1	114.96	114.72	189.76	0.00248
Observations	1,576	1,576	1,866	107
R^2	0.14469	0.14540	0.13997	-315.06

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

5.4 Heterogeneous Impacts

[Heterogeneous impacts placeholder: by urban/rural, by household income quartile, by age group, by parental education.]

5.5 Robustness to Alternative Explanations

[Robustness placeholder.]

6 Discussion

[Discussion placeholder.]

6.1 Implications for Policy and Research

[Policy implications placeholder.]

6.2 Conclusion

[Conclusion placeholder.]

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7 Appendix

7.1 Instrument Summary Statistics and First Stage Regressions

[Appendix placeholder.]

7.2 Shift-Share Instrument Diagnostics and Summary Statistics

[Appendix placeholder.]

7.3 Additional Regression Results: Contemporaneous Specification

Tables 10–15 replicate the main results using a contemporaneous specification in which development area tourism expenditures and the Bartik instrument enter without a lag.

Table 10: School Enrollment (Ages 5–19), IV Estimates (Contemporaneous)

Dependent Variable:	ENROLLED_IN_SCHOOL		
Model:	(1)	(2)	(3)
Variables			
Tourism exp. (DA, M USD)	-0.002	-0.002	0.020
	(0.004)	(0.004)	(0.017)
Female	0.006*	0.006*	0.028
	(0.003)	(0.003)	(0.023)
Age	-0.038***	-0.038***	-0.041***
	(0.0010)	(0.0010)	(0.004)
No. enrolled siblings (HH)		0.002	
		(0.004)	
Fixed-effects			
DEV_AREA	Yes	Yes	Yes
YEAR	Yes	Yes	Yes
Fit statistics			
F-test (1st stage), DA_EXP_	33.869	34.270	13.844
Observations	13,687	13,687	882
R^2	0.17705	0.17776	-0.13419

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 11: Reason for Non-Enrollment (Ages 5–19, Not Currently Enrolled), IV Estimates (Contemporaneous)

Dependent Variables:	DROPOUT_MONEY	DROPOUT_FAMILY	DROPOUT_NO_INTEREST	DROPOUT_PREGNANT
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp. (DA, M USD)	-0.004	-0.0008	-0.0001	0.003
	(0.003)	(0.002)	(0.002)	(0.003)
Female	-0.034***	-0.005	-0.041***	0.081***
	(0.013)	(0.009)	(0.009)	(0.011)
Age	-0.002	-0.004**	-0.005	-0.004*
	(0.003)	(0.002)	(0.003)	(0.002)
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_	5.5600	5.5600	5.5600	5.5600
Observations	1,655	1,655	1,655	1,655
R^2	-0.35754	-0.00468	0.04308	-0.24539

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 12: School Attendance Levels (Enrolled Sample), IV Estimates (Contemporaneous)

Dependent Variables:	SCHOOL_DAYS_SENT_PER_WEEK		KEPTFROMSCHOOL_ANY	
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp. (DA, M USD)	0.011	0.010	0.001	-0.013
	(0.043)	(0.042)	(0.004)	(0.013)
Female	0.052	0.052	0.001	0.040*
	(0.113)	(0.113)	(0.005)	(0.023)
Age	-0.026**	-0.026**	-0.007***	-0.009**
	(0.012)	(0.011)	(0.001)	(0.004)
No. enrolled siblings (HH)		-0.044		
		(0.079)		
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_	27.349	27.989	28.599	15.858
Observations	11,557	11,557	12,032	774
R^2	0.02590	0.02671	0.21877	0.06317

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 13: Reasons for School Absence (Occasional or Frequent, Annual; Enrolled Sample), IV Estimates (Contemporaneous)

Dependent Variables:	SCHOOL_KEPTYEAR_ANY_OCC_OR_FREQ	SCHOOL_KEPTYEAR_MONEY_OCC_OR_FREQ	SCHOOL_KEPTYEAR_TRANSPORT_OCC_OR_FREQ	SCHOOL_KEPTYEAR_ILL_OCC_OR_FREQ	SCHOOL_KEPTYEAR_WORK_OCC_OR_FREQ	SCHOOL_KEPTYEAR_MONEY_OCC_OR_FREQ
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp. (DA, M USD)	-0.003	-0.001	0.0008	-0.003	-4.01×10^{-5}	-0.007
	(0.004)	(0.004)	(0.0009)	(0.002)	(0.0001)	(0.014)
Female	-0.027***	-0.018***	-0.0001	-0.002	-0.0004	-0.023
	(0.007)	(0.007)	(0.001)	(0.004)	(0.0004)	(0.033)
Age	-0.002	0.001	0.0005***	-0.004***	3.1×10^{-5}	0.004
	(0.001)	(0.001)	(0.0002)	(0.0006)	(5.74×10^{-5})	(0.006)
Fixed-effects						
DEV_AREA	Yes	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
F-test (1st stage), DA_EXP_	28.599	28.599	28.599	28.599	28.599	15.858
Observations	12,032	12,032	12,032	12,032	12,032	774
R^2	0.02657	0.02853	-0.31619	-0.14064	0.00075	0.22849

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 14: School Expenditure by Category, Level (Enrolled Sample, Real 2024 USD), IV Estimates (Contemporaneous)

Dependent Variables:	SCHOOL_TUITION_NOBOOKS	SCHOOL_BOOKS_EXPENSES	SCHOOL_TRANSPORT_EXPENSES	SCHOOL_UNIFORM_EXPENSES	SCHOOL_LUNCHSNACKS_EXPENSES
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp. (DA, M USD)	-0.561	-1.46	0.696	-1.25	-0.308
	(0.972)	(1.64)	(1.33)	(1.28)	(2.06)
Female	-3.36	6.22***	24.8***	-7.50***	15.2**
	(6.06)	(1.72)	(7.90)	(1.81)	(6.70)
Age	-7.19***	1.06***	31.6***	2.63***	29.2***
	(0.803)	(0.247)	(2.52)	(0.272)	(1.24)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_	27.475	18.173	21.242	20.497	19.198
Observations	9,384	10,427	10,429	10,448	10,483
R^2	0.03268	-0.46742	0.07480	-0.15466	0.11723

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 15: School Expenditure by Category, Log Outcomes (Enrolled Sample), IV Estimates (Contemporaneous)

Dependent Variables:	LOG_SCHOOL_TUITION_NOBOOKS	LOG_SCHOOL_BOOKS_EXPENSES	LOG_SCHOOL_TRANSPORT_EXPENSES	LOG_SCHOOL_UNIFORM_EXPENSES	LOG_SCHOOL_LUNCHSNACKS_EXPENSES
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp. (DA, M USD)	-0.051	-0.073	0.072	0.0005	0.0005
	(0.113)	(0.104)	(0.083)	(0.031)	(0.040)
Female	0.047	0.684***	0.897***	0.083	-0.072
	(0.188)	(0.197)	(0.236)	(0.140)	(0.111)
Age	-0.548***	-0.331***	1.12***	0.021	0.180***
	(0.034)	(0.037)	(0.041)	(0.020)	(0.019)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_	27.475	18.173	21.242	20.497	19.198
Observations	9,384	10,427	10,429	10,448	10,483
R^2	0.02439	-0.00409	0.07734	0.02010	0.02957

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 16: Examination Attainment: At Least Level X (Ages 5–19), IV Estimates (Contemporaneous)

Dependent Variables:	EXAM_ATLEAST_GNAT	EXAM_ATLEAST_CXC_BASIC	EXAM_ATLEAST_CXC_GEN		MATHENG_BOTH
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp. (DA, M USD)	0.001	0.0009	0.0006	-0.002	6.85×10^{-5}
	(0.002)	(0.002)	(0.001)	(0.002)	(0.0007)
Female	0.032***	0.031***	0.026***	0.014	0.011***
	(0.004)	(0.004)	(0.004)	(0.011)	(0.002)
Age	0.021***	0.019***	0.016***	0.008***	0.007***
	(0.0008)	(0.0009)	(0.0009)	(0.002)	(0.0008)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_	33.869	33.869	33.869	13.844	33.869
Observations	13,687	13,687	13,687	882	13,687
R^2	0.11338	0.11290	0.10256	0.07354	0.05165

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 17: Upper Secondary and Tertiary Attainment (Ages 14–25), IV Estimates (Contemporaneous)

Dependent Variables:	EXAM_ATLEAST_ALEVEL	EXAM_ATLEAST_DEGREE	CURRENTLY_IN_COLLEGE		
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp. (DA, M USD)	-0.001	8.52×10^{-5}	-0.0006	-0.0008	0.0007
	(0.001)	(0.0002)	(0.0009)	(0.001)	(0.002)
Female	0.024***	0.008***	0.029***	0.028***	0.023*
	(0.004)	(0.002)	(0.005)	(0.005)	(0.012)
Age	0.009***	0.003***	0.005***	0.006***	0.002
	(0.001)	(0.0006)	(0.0007)	(0.001)	(0.001)
No. enrolled siblings (HH)				-0.004*	
				(0.003)	
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_	34.836	34.836	34.836	27.184	6.2465
Observations	9,813	9,813	9,813	8,209	441
R^2	-0.02153	0.01979	0.02766	0.02382	0.19838

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 18: Skills Program Enrollment and Completion (Ages 5–19), IV Estimates (Contemporaneous)

Dependent Variables:	ENROLLED_SKILLED_PROGRAM		DIPLOMA_SKILLS_PROGRAM	
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp. (DA, M USD)	-0.0002	-0.0001	0.001	0.028
	(0.0004)	(0.0004)	(0.002)	(0.050)
Female	0.003	0.003	-0.003	-0.012
	(0.002)	(0.002)	(0.011)	(0.124)
Age	-0.0006	-0.0006	0.015***	0.003
	(0.0006)	(0.0006)	(0.004)	(0.030)
No. enrolled siblings (HH)		-0.001*		
		(0.0006)		
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_	3.6472	3.5134	5.7144	1.3493
Observations	1,594	1,594	1,886	109
R^2	0.12931	0.13597	0.12881	-0.32171

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.4 Additional Regression Results: Two-Year Lagged Specification

Tables 19–24 replicate the main results using a two-year lag specification in which development area tourism expenditures and the Bartik instrument enter with a two-year lag.

Table 19: School Enrollment (Ages 5–19), IV Estimates (Lag 2)

Dependent Variable:	ENROLLED_IN_SCHOOL		
Model:	(1)	(2)	(3)
Variables			
Tourism exp., lag 2 (DA, M USD)	-0.0004	-0.0004	-0.001
	(0.0007)	(0.0007)	(0.003)
Female	0.006*	0.006*	0.015
	(0.003)	(0.003)	(0.024)
Age	-0.038***	-0.038***	-0.043***
	(0.001)	(0.001)	(0.003)
No. enrolled siblings (HH)		-0.0001	
		(0.002)	
Fixed-effects			
DEV_AREA	Yes	Yes	Yes
YEAR	Yes	Yes	Yes
Fit statistics			
F-test (1st stage), DA_EXP_lag2	502.58	502.45	28.186
Observations	13,518	13,518	861
R^2	0.25498	0.25499	0.32155

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 20: Reason for Non-Enrollment (Ages 5–19, Not Currently Enrolled), IV Estimates (Lag 2)

Dependent Variables:	DROPOUT_MONEY	DROPOUT_FAMILY	DROPOUT_NO_INTEREST	DROPOUT_PREGNANT
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 2 (DA, M USD)	0.0002	0.0007	0.001	-0.0010
	(0.0005)	(0.0007)	(0.002)	(0.0010)
Female	-0.031***	-0.004	-0.039***	0.079***
	(0.011)	(0.009)	(0.010)	(0.011)
Age	-0.004	-0.005**	-0.005	-0.003
	(0.004)	(0.002)	(0.003)	(0.002)
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_lag2	66.769	66.769	66.769	66.769
Observations	1,637	1,637	1,637	1,637
R^2	0.08550	0.03685	0.01678	0.05667

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 21: School Attendance Levels (Enrolled Sample), IV Estimates (Lag 2)

Dependent Variables:	SCHOOL_DAYS_SENT_PER_WEEK		KEPTFROMSCHOOL_ANY	
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 2 (DA, M USD)	-0.001	-0.001	0.0010*	0.012
	(0.015)	(0.015)	(0.0006)	(0.009)
Female	0.035	0.035	0.0009	0.033
	(0.113)	(0.113)	(0.005)	(0.028)
Age	-0.029**	-0.029***	-0.007***	-0.008*
	(0.011)	(0.011)	(0.0008)	(0.004)
No. enrolled siblings (HH)		-0.035		
		(0.078)		
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_lag2	434.69	434.51	439.38	18.957
Observations	11,412	11,412	11,881	756
R^2	0.03440	0.03447	0.22596	0.11498

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 22: Reasons for School Absence (Occasional or Frequent, Annual; Enrolled Sample), IV Estimates (Lag 2)

Dependent Variables:	SCHOOL_KEPTYEAR_ANY_OCC_OR_FREQ	SCHOOL_KEPTYEAR_MONEY_OCC_OR_FREQ	SCHOOL_KEPTYEAR_TRANSPORT_OCC_OR_FREQ	SCHOOL_KEPTYEAR_ILL_OCC_OR_FREQ	SCHOOL_KEPTYEAR_WORK_OCC_OR_FREQ	SCHOOL_KEPTYEAR_MONEY_OCC_OR_FREQ
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp., lag 2 (DA, M USD)	0.001	-0.002	-4.46×10^{-5}	0.004	-8.32×10^{-6}	0.020
	(0.002)	(0.001)	(0.0002)	(0.003)	(1.81×10^{-5})	(0.029)
Female	-0.026***	-0.017**	-5.64×10^{-5}	-0.003	-0.0004	-0.051
	(0.006)	(0.007)	(0.001)	(0.006)	(0.0004)	(0.039)
Age	-0.002	0.001	0.0004***	-0.004***	3.59×10^{-5}	0.007
	(0.001)	(0.0009)	(0.0001)	(0.0006)	(5.32×10^{-5})	(0.006)
Fixed-effects						
DEV_AREA	Yes	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
F-test (1st stage), DA_EXP_lag2	439.38	439.38	439.38	439.38	439.38	18.957
Observations	11,881	11,881	11,881	11,881	11,881	756
R^2	0.13048	0.05276	0.01873	-0.04748	0.00630	-0.21224

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 23: School Expenditure by Category, Level (Enrolled Sample, Real 2024 USD), IV Estimates (Lag 2)

Dependent Variables:	SCHOOL_TUITION_NOBOOKS	SCHOOL_BOOKS_EXPENSES	SCHOOL_TRANSPORT_EXPENSES	SCHOOL_UNIFORM_EXPENSES	SCHOOL_LUNCHSNACKS_EXPENSES
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 2 (DA, M USD)	0.163	-0.148	-0.398	0.091	0.935
	(0.370)	(0.128)	(0.697)	(0.229)	(0.899)
Female	-4.24	5.61***	25.7***	-8.41***	14.9**
	(6.72)	(1.90)	(7.98)	(1.88)	(6.54)
Age	-7.23***	1.17***	31.9***	2.71***	29.4***
	(0.774)	(0.246)	(2.48)	(0.275)	(1.27)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag2	993.40	811.13	773.80	810.83	804.73
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.03981	0.06848	0.08087	0.04630	0.11294

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 24: School Expenditure by Category, Log Outcomes (Enrolled Sample), IV Estimates (Lag 2)

Dependent Variables:	LOG_SCHOOL_TUITION_NOBOOKS	LOG_SCHOOL_BOOKS_EXPENSES	LOG_SCHOOL_TRANSPORT_EXPENSES	LOG_SCHOOL_UNIFORM_EXPENSES	LOG_SCHOOL_LUNCHSNACKS_EXPENSES
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 2 (DA, M USD)	-0.004	-0.020	-0.0007	0.0008	-0.009
	(0.013)	(0.022)	(0.021)	(0.014)	(0.014)
Female	-0.004	0.660***	0.997***	0.053	-0.074
	(0.146)	(0.213)	(0.224)	(0.137)	(0.109)
Age	-0.550***	-0.320***	1.12***	0.025	0.183***
	(0.036)	(0.036)	(0.038)	(0.019)	(0.019)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag2	993.40	811.13	773.80	810.83	804.73
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.08555	0.06388	0.13996	0.02048	0.02880

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 25: Examination Attainment: At Least Level X (Ages 5–19), IV Estimates (Lag 2)

Dependent Variables:	EXAM_ATLEAST_GNAT	EXAM_ATLEAST_CXC_BASIC	EXAM_ATLEAST_CXC_GEN		MATHENG_BOTH
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 2 (DA, M USD)	-1.71×10^{-6}	-5.65×10^{-6}	-0.0002	0.0003	-0.0002
	(0.0004)	(0.0003)	(0.0004)	(0.001)	(0.0002)
Female	0.031***	0.030***	0.025***	0.018*	0.011***
	(0.004)	(0.004)	(0.004)	(0.011)	(0.002)
Age	0.021***	0.019***	0.016***	0.008***	0.007***
	(0.0008)	(0.0009)	(0.0010)	(0.002)	(0.0008)
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag2	502.58	502.58	502.58	28.186	502.58
Observations	13,518	13,518	13,518	861	13,518
R^2	0.14505	0.13747	0.11735	0.10646	0.05172

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 26: Upper Secondary and Tertiary Attainment (Ages 14–25), IV Estimates (Lag 2)

Dependent Variables:	EXAM_ATLEAST_ALEVEL	EXAM_ATLEAST_DEGREE	CURRENTLY_IN_COLLEGE		
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 2 (DA, M USD)	-0.0002	-0.0001	-0.0002	-0.0004	2.44×10^{-5}
	(0.0004)	(8.84×10^{-5})	(0.0003)	(0.0005)	(0.0009)
Female	0.023***	0.007***	0.029***	0.028***	0.022*
	(0.004)	(0.002)	(0.004)	(0.005)	(0.012)
Age	0.009***	0.003***	0.005***	0.006***	0.002
	(0.001)	(0.0006)	(0.0007)	(0.0009)	(0.002)
No. enrolled siblings (HH)				-0.005**	
				(0.002)	
Fixed-effects					
DEV_AREA	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes
Fit statistics					
F-test (1st stage), DA_EXP_lag2	368.80	368.80	368.80	302.88	10.696
Observations	9,706	9,706	9,706	8,116	429
R^2	0.05391	0.02513	0.03284	0.03432	0.19959

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 27: Skills Program Enrollment and Completion (Ages 5–19), IV Estimates (Lag 2)

Dependent Variables:	ENROLLED_SKILLED_PROGRAM		DIPLOMA_SKILLS_PROGRAM	
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 2 (DA, M USD)	-0.0003	-0.0003	-0.002	-0.0005
	(0.001)	(0.001)	(0.003)	(0.005)
Female	0.003	0.003	-0.004	-0.073
	(0.003)	(0.002)	(0.012)	(0.049)
Age	-0.0004	-0.0005	0.017***	0.013
	(0.001)	(0.001)	(0.004)	(0.015)
No. enrolled siblings (HH)		-0.001**		
		(0.0005)		
Fixed-effects				
DEV_AREA	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), DA_EXP_lag2	9.2335	9.2371	14.644	7.4653
Observations	1,576	1,576	1,866	107
R^2	0.13100	0.13116	0.11263	0.46376

Clustered (DEV_AREA) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.5 Tourism Summary Statistics

[Appendix placeholder.]

7.6 Additional Household and Education Summary Statistics

[Appendix placeholder.]